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Hygiene management device for entrance hall and control method of same

Abstract

A hygiene management device for an entrance hall and a control method of the same are proposed. The hygiene management device may be installed in the space of the entrance hall, and when the opening of an entrance door is detected, air is discharged toward the entrance door by tilting an air discharger toward the entrance door, and when the closing of the entrance door is detected, the air discharger is tilted to a preset initial position to stop the discharging of air. Accordingly, during the opening of the entrance door, the introduction of external air into the space of the entrance hall may be minimized.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) The present application claims the priority benefit of Korean Patent Application No. 10-2020-0174512 filed on Dec. 14, 2020 in the Republic of Korea and 10-2021-0015906 filed on Feb. 4, 2021, in the Republic of Korea, which are hereby incorporated by reference in their entirety for all purposes as if fully set forth herein.

BACKGROUND OF THE INVENTION

1. Field of the Invention

(2) The present disclosure relates to a hygiene management device for an entrance hall and a control method of the same.

2. Description of the Background Art

(3) Generally, in order to enter an indoor space from the outside, a visitor is required to go through an entrance hall which separates the outside from the indoor space. For example, in the case of a house, a vestibule as an entrance hall is provided, and thus to enter the house, a person is required to go through the vestibule.

(4) Recently, air quality outside is getting worse due to yellow dust or fine dust. In this situation, when yellow dust or fine dust enters an indoor space from the outside, the indoor space may be contaminated.

(5) In a dense environment (i.e., a small indoor space), such as an apartment, removing contaminants outside an entrance door may cause the contaminants to be transferred to surrounding people. Accordingly, before a visitor enters an indoor space, it is necessary to remove contaminants attached to the visitor in an entrance hall such as a vestibule.

(6) Technologies for preventing contaminants from being transferred to the indoor space are disclosed.

(7) An air shower for a vestibule of a house is disclosed in (Patent Document 1) Korean Patent Application Publication No. 10-2009-0040630. In the related art disclosed in (Patent Document 1), air is showered on a visitor to remove contaminants, such as dust, on the clothing of the visitor.

(8) However, the air applied to the visitor flows downward from the upper portion, so the air is applied mainly to the visitor's head and to the upper portion of the visitor's upper body and thus contaminants on the visitor's lower body and shoes are not properly removed. Furthermore, in the vestibule, an air introduction hole into, which air is introduced, is located at the side surface of a

vestibule space, so contaminants such as dust remain on the bottom of the vestibule.

(9) An air shower system for a vestibule is disclosed in (Patent Document 2) Korean Patent Application Publication No. 10-2020-0117286. In the air shower system disclosed in (Patent Document 2), in addition to the air shower, a photocatalytic lamp is installed on a ceiling to simultaneously remove contaminants, such as dust, and harmful substances, such as viruses, on the clothing of a visitor.

(10) However, even the related art disclosed in (Patent Document 2), has the problems that contaminants, such as dust, remain on the bottom of a vestibule, and contaminants on a visitor's lower body and shoes are not properly removed due to the emission of light by the photocatalytic lamp from the ceiling.

(11) A lighting-integrated air cleaning system for entrance space and a control method of the same are disclosed in (Patent Document 3) Korean Patent Application Publication No. 10-2020-0046715. In the related art disclosed in (Patent Document 3), a lighting fixture and an air cleaning system are integrated with each other such that contaminants are removed from a visitor in an entrance space such as a vestibule.

(12) However, even in the related art disclosed in (Patent Document 3), a structure in which air is simply discharged downward from the upper side is integrated with the light fixture, and an air introduction hole is formed in the side surface of the entrance space, so contaminants in the entrance space are not properly removed.

(13) An air shower apparatus integrated with a vestibule door is disclosed in (Patent Document 4) Korean Patent Application Publication No. 10-2019-0055303. In the related art disclosed in (Patent Document 4), a shower booth having a shower space thereinside is separately installed, such that a visitor can remove contaminants in the shower booth.

(14) However, the shower booth disclosed in (Patent Document 4) is separately partitioned in space inside a vestibule, and thus a space occupied by an entire device inside the vestibule increases, so the air shower apparatus is not effective. Furthermore, the air shower apparatus has a simple structure in which air is discharged downward from an upper side, so ability with which the air shower apparatus removes contaminants decreases.

(15) In addition in the related arts disclosed in Patent Documents 1 to 4 described above, in a process in which a visitor opens and closes an entrance door, outdoor contaminants are introduced through the entrance door into a vestibule and the internal space of the vestibule is contaminated.

Documents of Related Art

(16) (Patent Document 1) Korean Patent Application Publication No. 10-2009-0040630 (Patent Document 2) Korean Patent Application Publication No. 10-2020-0117286 (Patent Document 3) Korean Patent Application Publication No. 10-2020-0046715 (Patent Document 4) Korean Patent Application Publication No. 10-2019-0055303

SUMMARY OF THE INVENTION

(17) Accordingly, the present disclosure has been made keeping in mind the above problems occurring in the related art, and the present disclosure is intended to propose a hygiene management device for an entrance hall and the control method of the same in which air is discharged to a user, such that contaminants attached to the user's body and clothes are removed.

(18) The present disclosure is intended to propose a hygiene management device for an entrance hall and the control method of the same involving a process in which a user opens and closes an entrance door to go in and out of an indoor area, external contaminants are prevented from being introduced into the entrance hall.

(19) The present disclosure is intended to propose a hygiene management device for an entrance hall and the control method of the same in which after contaminated air in the entrance hall is introduced into a footrest to be purified, the purified air is discharged back into the entrance hall.

(20) The present disclosure is intended to propose a hygiene management device for an entrance hall and the control method of the same in which a user going into the entrance hall is detected and

air is automatically discharged to the user.

(21) The present disclosure is intended to propose a hygiene management device for an entrance hall and the control method of the same in which when a user is located at a preset specific position within the hygiene management device, the user is detected and air is automatically discharged to the user.

(22) The present disclosure is intended to propose a hygiene management device for an entrance hall and the control method of the same in which the discharge direction of air may be changed according to the opening and closing of the entrance door.

(23) The present disclosure is intended to propose a hygiene management device for an entrance hall and the control method of the same in which the discharge direction of air is changed according to a position to which a user moves.

(24) According to the hygiene management device for an entrance hall of the present disclosure and the control method of the same, an air discharger for discharging air inside a specific space may be provided.

(25) According to the hygiene management device for an entrance hall of the present disclosure and the control method of the same, the air discharger may discharge air to a user located inside a specific space, such that contaminants on the user's clothing or shoes can be removed.

(26) According to the hygiene management device for an entrance hall of the present disclosure and the control method of the same, when a user stands on a footrest, the user may be detected and the air discharger may discharge air toward the user.

(27) According to the hygiene management device for an entrance hall of the present disclosure and the control method of the same, when an entrance door is opened, the air discharger may be tilted toward the entrance door to discharge air. Accordingly, external air may be prevented from being introduced through the opened entrance door into the entrance hall from the outside.

(28) The hygiene management device for an entrance hall of the present disclosure may include: a first detector configured to detect the opening/closing of the entrance door installed in the entrance hall, an air discharger mounted to the upper portion of the entrance hall and configured to discharge air, a fan assembly configured to supply a blowing force such that air is discharged by the air discharger, a lift assembly configured to tilt the air discharger, and a controller configured to drive the fan assembly and the lift assembly, wherein when the first detector detects the opening of the entrance door, the controller may drive the lift assembly to tilt the air discharger and may drive the fan assembly such that the tilted air discharger discharges air.

(29) The hygiene management device for an entrance hall of the present disclosure may further include a second detector configured to detect whether a user is present on a footrest of the hygiene management device, wherein when the user is detected by the second detector, the controller may drive the fan assembly such that the air discharger discharges air toward the user.

(30) According to the hygiene management device for an entrance hall of the present disclosure and the control method of the same, the lift assembly may tilt the air discharger such that the direction of air discharged by the air discharger is directed toward an entrance door.

(31) According to the hygiene management device for an entrance hall of the present disclosure and the control method of the same, when the entrance door is opened, the lift assembly may be driven such that the air discharger is tilted toward the entrance door, and air may be discharged toward the entrance door, and later when the entrance door is closed, the lift assembly may be driven to tilt the air discharger to a preset initial position and the air discharge may stop.

(32) According to the hygiene management device for an entrance hall of the present disclosure and the control method of the same, when tilting the air discharger to an initial position, the direction of air discharged by the air discharger may be directed toward a center of the space of the entrance hall.

(33) According to the hygiene management device for an entrance hall of the present disclosure and the control method of the same, when a user stands on the footrest, the fan assembly may operate

such that the internal air of an entrance space is introduced into the footrest through multiple introduction holes formed in the footrest.

(34) According to the hygiene management device for an entrance hall of the present disclosure and the control method of the same, air introduced into the footrest may pass through one or multiple filters and a duct, and may be discharged back into the entrance space through the air discharger.

(35) According to the hygiene management device for an entrance hall of the present disclosure and the control method of the same, when a preset period of time elapses after air is discharged toward a user on the footrest, the air discharger may stop air discharge.

(36) According to the hygiene management device for an entrance hall of the present disclosure and the control method of the same, even before the preset period of time elapses, when a user is not detected on the footrest, the air discharger may stop air discharge.

(37) According to the hygiene management device for an entrance hall of the present disclosure, the lift assembly may include first and second lifts installed on the opposite edges of the air discharger, respectively, and lift motors configured to drive the first and second lifts, respectively. When tilting the air discharger, one of the first lift and the second lift may operate to tilt the air discharger such that an edge of a side of the air discharger is lowered.

(38) According to the hygiene management device for an entrance hall of the present disclosure, multiple vanes may be disposed at predetermined intervals in the air discharger so as to preset the direction of air discharged by the air discharger (i.e., air discharged by the air discharger follows an air flow path defined by the multiple vanes), wherein each of the vanes may have height increasing gradually toward the flow direction of air.

(39) The control method of a hygiene management device for an entrance hall of the present disclosure may include: detecting whether the entrance door mounted to the entrance hall is opened, tilting the air discharger toward the entrance door when the opening is detected, and discharging air by the air discharger.

(40) The control method of a hygiene management device for an entrance hall of the present disclosure may further include stopping the discharging of air by the air discharger such that after the discharging of air, whether the entrance door is closed is detected and when the closing of the entrance door is detected, the air discharger is tilted to a preset initial position.

(41) The hygiene management device for an entrance hall according to the present disclosure and the control method of the same may have at least one of the following effects.

(42) According to the present disclosure, air may be discharged to a user, thereby removing contaminants attached to a user's body and clothes.

(43) According to the present disclosure, in the entrance space blocked from an indoor area, air may be discharged toward a user, thereby preventing external contaminants from being introduced into the indoor area.

(44) According to the present disclosure, in a process in which a user opens and closes the entrance door to go in and out of the indoor area, external contaminants may be prevented from being introduced into the entrance hall.

(45) According to the present disclosure, air in the space of the entrance hall may be introduced into the footrest and be purified, and the purified air may be discharged back into the space of the entrance hall, thereby managing the inside of the space of the entrance hall to be pleasant.

(46) According to the present disclosure, when a user stands on the footrest which is designated, air may be automatically discharged toward a user, thereby providing convenience.

(47) According to the present disclosure, the discharge direction of air may be changed according to the opening and closing of the entrance door, thereby preventing the introduction of external air into the entrance space and removing the internal contaminants of the entrance space in correspondence to the opening and closing of the entrance door.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The above and other objectives, features, and other advantages of the present disclosure will be more clearly understood from the following detailed description when taken in conjunction with the accompanying drawings, in which:
- (2) FIG. 1 is a perspective view unidirectionally illustrating the configuration of a hygiene management device for an entrance hall according to an embodiment of the present invention.
- (3) FIG. 2 is a rear perspective view illustrating the configuration of the hygiene management device according to an embodiment of the present invention.
- (4) FIG. 3 is a sectional view illustrating the configuration of the hygiene management device according to an embodiment of the present invention.
- (5) FIG. 4 is a view illustrating the hygiene management device disposed in an entrance space according to an embodiment of the present invention.
- (6) FIG. 5 is a sectional view illustrating the internal configuration of an air management module according to an embodiment of the present invention.
- (7) FIG. 6 is a sectional perspective view illustrating the configuration of each of an air discharger and a second outlet of a duct according to an embodiment of the present invention.
- (8) FIG. 7 is a sectional view illustrating the configuration of each of portions of the air discharger and the duct in which vanes are installed and the flow of air according to an embodiment of the present invention.
- (9) FIGS. 8A and 8B are side views illustrating the configuration of an important part of each of the vanes illustrated in FIG. 7.
- (10) FIGS. 9A and 9B are views illustrating first and second lifts in the air discharger according to an embodiment of the present invention.
- (11) FIG. 10 is a block diagram of components connected to a controller according to an embodiment of the present invention.
- (12) FIGS. 11 to 14 are flowcharts illustrating the control method of a hygiene management device for an entrance hall according to embodiments of the present invention.
- (13) FIG. 15 is a view illustrating that the discharge and introduction of air are performed when a user is located on a footrest according to an embodiment of the present invention.
- (14) FIG. 16 is a view illustrating a state in which the air discharger is tilted toward the entrance door when a user comes into the entrance space through an entrance door according to an embodiment of the present invention.
- (15) FIG. 17 is a view illustrating a state in which the air discharger is tilted toward the entrance door when a user goes out of the entrance space through the entrance door according to an embodiment of the present invention.
- (16) FIG. 18A is a view illustrating a state in which air can be discharged toward the center of the lower portion of the entrance space, and FIG. 18B is a view illustrating a state in which air can be discharged toward the lower portion of the entrance space.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

- (17) The exemplary embodiment of a hygiene management device for an entrance hall of the present disclosure and the control method of the same will be described in detail with reference to the accompanying drawings.
- (18) The hygiene management device **1000** according to the embodiment of the present disclosure may be installed inside the space **12** (e.g., entrance space **12**) of the entrance hall **10** for access to a specific indoor space. When the hygiene management device is installed in a residential space **12**, the hygiene management device may be installed in a vestibule. When the hygiene management device is installed in an office space **12**, the hygiene management device may be installed in space

which is separately partitioned in the entrance hall of the associated office space.

(19) An entrance door **14** for going out to or coming in from the outside may be provided at a first side surface of the entrance hall **10**, and an inside door **16** for entering and exiting the indoor space may be provided at a second side surface of the entrance hall **10** opposite to the first side surface of the entrance hall **10**.

(20) A footrest **100** may be installed on the bottom of the entrance hall **10** having space **12** defined therein. The footrest **100** may have a predetermined area (i.e., a predetermined size and shape) such that a visitor can stand on the footrest **100** and may function to support the visitor.

(21) As illustrated in the drawing, the footrest **100** may be configured as a rectangular plate shape having space defined therein. However, according to the shape of the entrance space **12**, the shape of the footrest **100** may be configured to have various plate shapes or any type of shape.

(22) A plurality of introduction holes may be formed in the upper surface of the footrest **100** such that air in the space **12** of the entrance hall **10** is introduced into the footrest **100**, and due to the operation of a fan assembly **300**, which is described below, the air in the space **12** of the entrance hall **10** may be introduced into the footrest **100**.

(23) The air introduced into the internal space of the footrest **100** through such introduction holes may be transferred through a predetermined flow path to an air management module **200**. The air management module **200** may remove foreign matter contained in the air transferred thereto through the predetermined flow path and may function to manage the humidity and temperature of the air.

(24) The air management module **200** may include a plurality of filters **221**, **223**, and **225**. These filters **221**, **223**, and **225** may be disposed in an air flow path formed in the air management module such that air supplied to the air management module **200** passes through the filters **221**, **223**, and **225**.

(25) Each of the filters **221**, **223**, and **225** may have a specific function. For example, a first filter **221** may have the function of removing moisture. A second filter **223** may have the function of removing contaminants and dust (e.g., fine dust). A third filter **225** may have an antibacterial function. Alternatively, filters having other functions may be included in the air management module **200**.

(26) In order to preset the temperature of air passing through the air management module **200**, a heat exchanger **240** may be installed. The heat exchanger **240** may exchange heat with air passing through the filters **221**, **223**, and **225**. The heat exchanger **240** may be installed in a heat exchanger casing **242**.

(27) Air may flow through the heat exchanger casing **242** in a front-to-rear direction, with the front direction facing towards the footrest **100** and the rear direction facing away from the footrest **100**. The outlet of the heat exchanger casing **242** may communicate with the fan assembly **300** to be described below. There may be components of a heat exchange cycle for heat exchange. Reference numeral **244** denotes a compressor.

(28) The fan assembly **300** may be mounted to the rear surface of the air management module **200**, that is, the outlet of the air management module **200** through which air flows out. The fan assembly **300** may supply an air blowing force such that air flows in the hygiene management device **1000**. Due to such an air blowing force, an air discharger **500** to be described below may discharge air to the outside.

(29) A fan housing **310** may constitute the exterior of the fan assembly **300**. A housing internal space **314** may be defined in the fan housing **310**, and the housing internal space **314** may be configured such that air introduced to the fan housing **310** from a position in front of the fan housing **310** is discharged from the fan housing **310** through an outer surface thereof.

(30) A fan **316** may be installed in the fan housing **310**. The fan **316** may be driven by a fan motor to allow air to be introduced to the fan housing **310** through an inlet of the fan housing **310** and to allow the air to be discharged through the outlet of the fan housing **310**.

- (31) A function module **350** may be installed on a position corresponding to the upper side of the air management module **200** (i.e., positioned above the air management module **200**). There may be a clothing management device as an example of the function module **350**. When clothes are placed inside the clothing management device, the clothing management device can remove dust from the clothes or sterilize the clothes. In addition, the clothing management device may function to unwrinkle clothing, such as by steaming. Further, any type of item may be inserted into the function module **350**, such as shoes, glasses, an umbrella or the like.
- (32) As other examples of the function module **350**, there may be various devices such as a shoe management device, an umbrella management device, a shoe storage device, and an umbrella storage device. For reference, each of these function modules **350** may not be configured as an independent function module, but some of the function modules may be combined with each other.
- (33) A control panel **352** may be mounted to the front surface part of the function module **350** (i.e., a surface disposed between the footrest **100** and the air discharger **500**) so as to preset and control various operations and functions of the hygiene management device **1000** of the present disclosure. Such a control panel **352** may preset and control the various operations and functions of the function module **350**.
- (34) The control panel **352** may be a device in which an input device and a display device (a display, such as a touch screen, and OLED screen and the like) are integrated with each other. The input device may be embodied as a touch pad or a touch screen.
- (35) A duct **400** may function to move an air flow generated in the fan assembly **300** to the relatively upper portion of the entrance hall **10**. The duct **400** may include a first duct **410** and a second duct **420**.
- (36) The first duct **410** may extend along the side wall of the entrance hall **10**, may extend vertically and may define an outer side surface of the hygiene management device **1000**. A first inlet **412** of the first duct **410** may be connected to the outlet of the fan housing **310** of the fan assembly **300**, and thus air may be transferred through the outlet of the fan housing **310** to the first inlet **412** of the first duct **410**.
- (37) The first duct **410** may have a branch flow path through which air can be supplied to the function module **350**. There may be a separate damper inside each of the branch flow path and the first duct **410** to control the flow of air therein.
- (38) The second duct **420** may be located on the upper portion of the entrance hall **10**. The second duct **420** may allow air passing through the first duct **410** to flow therein such that the air can be supplied to the air discharger **500** to be described below. The second duct **420** may have a first outlet **422** and a second outlet **424**.
- (39) The first outlet **422** may be located in a relatively upstream portion of the second duct **420** relative to the flowing direction of air inside the second duct **420**, and the second outlet **424** may be located in a relatively downstream portion of the second duct **420** compared to the first outlet **422**. As illustrated in FIG. 3, the first outlet **422** may be located closer to the first duct **410** than the second outlet **424**.
- (40) The air discharger **500** may be connected to the second duct **420** of the upper portion of the entrance hall **10**. The air discharger **500** may discharge air transferred through the duct **400** into the space **12** of the entrance hall **10**.
- (41) The air discharger **500** may discharge air transferred through each of the first outlet **422** and the second outlet **424** of the second duct **420** into the space **12**. The air discharger **500** may preset and change the amount (i.e., volume), speed, and direction of the transferred air.
- (42) Discharge holes through which air is discharged may be formed in the air discharger **500**. In the embodiment, the discharge holes may be formed in positions, respectively, corresponding to the first outlet **422** and the second outlet **424** of the second duct **420**. The discharge holes may be open to the lower surface of the air discharger **500**.
- (43) A corrugated pipe **556** may be installed on a portion on which the second duct **420** and the air

discharger **500** are connected to each other to prevent air from leaking even when the air discharger **500** moves.

(44) The corrugated pipe **556** may be flexible, and thus although the air discharger **500** moves relative to the duct **400**, a connection state between the second duct **420** and the air discharger **500** may be maintained. The corrugated pipe **556** may connect the second duct **420** with the air discharger **500** in each of the first outlet **422** and the second outlet **424** of the second duct **420**. That is, the corrugated pipe **556** may be provided in plurality.

(45) The air discharger **500** may be provided with a lift assembly **520**. The lift assembly **520** may include a first lift **521** and a second lift **521'**. The lift assembly **520** may be driven by a controller **900**, which is described below.

(46) The first and second lifts **521** and **521'** are intended to tilt the air discharger **500** by moving up and down the opposite edges of the air discharger **500**.

(47) The first lift **521** and the second lift **521'** may be installed on the opposite edges of the air discharger **500**, respectively, and a specific portion of the air discharger **500** may be moved down by these lifts **521** and **521'** so as to tilt the air discharger **500**.

(48) Accordingly, a direction (e.g., an angle) to which the lower surface of the air discharger **500** is directed may be adjusted to adjust the discharge direction of air. A driving force for operating the first and second lifts **521** and **521'** may be supplied by at least one lift motor **525**. Specifically, there may be two lift motors **525**, as shown in FIGS. **9A** and **9B**, and each lift motor **525** may be for applying the driving force for a respective one of the first and second lifts **521** and **521'**.

(49) A portion of each of parts of the first and second lifts **521** and **521'** may be mounted to the second duct **420** or the upper portion of the entrance space **12**, and the remaining portion thereof may be mounted to the air discharger **500**. That is, a first portion or a first end of each of parts of the first and second lifts **521** and **521'** may be mounted to the second duct **420** or the upper portion of the entrance space **12**, and a second portion or a second end thereof may be mounted to the air discharger **500**. Due to a specific part of the air discharger **500** moved up and down by the lifts **521** and **521'**, a direction to which the lower surface of the air discharger **500** is directed may be adjusted.

(50) The first and second lifts **521** and **521'** may be replaced by other components which perform the same functions, as known in the art. In the hygiene management device of the present disclosure, any components which can tilt the air discharger **500** in a specific direction may be applied.

(51) Multiple vanes **560** and **560'** (e.g., a plurality of vanes **560**, and a plurality of vanes **560'**) may be installed respectively between the first outlet **422** of the duct **400** and a first side discharge hole of the air discharger **500** and between the second outlet **424** of the duct **400** and a second side discharge hole of the air discharger **500**.

(52) A vane rotation center member **558** may be mounted to each of the first outlet **422** and the second outlet **424** of the second duct **420**. Multiple vanes **560** or **560'** may be mounted to each vane rotation center member **558** such that the multiple vanes can rotate at predetermined angles. That is, the vane rotation center member **558** controls the rotation of the plurality of vanes **560**, **560'** of the respective side of the air discharger **500**.

(53) Each of the vanes **560** or **560'** (i.e., plurality of vanes may preset the direction of air to be discharged through the discharge holes of the air discharger **500**. The vane **560** or **560'** may include multiple vanes disposed at predetermined intervals.

(54) The vane **560** or **560'** may function to preset the direction of air to be discharged through the discharge holes. As illustrated in FIGS. **8A** and **8B**, in the vane **560** or **560'**, an upper flow part **561** and a lower flow part **561'** which are plates, respectively, having predetermined shapes may be connected to each other such that the upper flow part **561** and the lower flow part **561'** have a predetermined angle therebetween. The upper flow part **561** and the lower flow part **561'** may be connected to each other by a connection curved part **561''** having a predetermined radius of

curvature.

(55) At least a portion of the upper flow part of the vane **560** or **560'** may be located inside the second duct **420**. The lower flow part of the vane **560** or **560'** may be mainly located inside the air discharger **500**.

(56) The vane **560** or **560'** may have more portions located in the second duct **420** gradually in an air flow direction in the second duct **420**. That is, the vane **560** or **560'** may be located at the position of height increasing gradually in the air flow direction in the second duct **420**.

Additionally, the vane **560** located in the second outlet **424** may be located at a position higher than the vane **560'** located in the first outlet **422**.

(57) Such configuration can be seen even in FIG. **3**. This is intended to allow air flowing in the second duct **420** to evenly flow through the entire portion of each of the first outlet **422** and the second outlet **424**.

(58) As for the length of the vane **560** or **560'**, the vane **560'** located at the first outlet **422** may be longer than the vane **560** located at the second outlet **424**. This is intended to allow air to flow through the discharge holes without air being wasted gradually toward the end of the second duct **420**.

(59) The vane **560** or **560'** may guide air to the lower portion of the air discharger **500**. According to the rotated state of the vane **560** or **560'** relative to the vane rotation center member **558**, a direction in which the vane guides air may be changed.

(60) A vane drive member **562** may be provided to control the operation state of each of the vane **560** and **560'**. Like the vane rotation center member **558**, the vane drive member **562** may be connected to each of the vanes **560** and **560'**.

(61) The vane drive member **562** may be connected relatively rotatably to the vane **560** or **560'** at a position located apart from a portion at which the vane rotation center member **558** is connected to the vane **560** or **560'** and toward the discharge holes. One vane drive member **562** may be used for each of the opposite sides of the vane **560** or **560'**. That is, one vane drive member **562** may be used for both of the vanes **560**, **560'**. Alternatively, only one vane drive member **562** may be used for the vane **560** or **560'**. That is, each vane **560** or **560'** may have its own vane drive member **562**.

(62) Vane motors **570** may be provided to drive the vanes **560** and **560'**, respectively. A drive link **574** may be connected to each of the vane motors **570**. The drive link **574** may be driven by the operation of the vane motor **570**. Further, each drive link **574** may be connected to the vane motor **570** via a drive part **580**. That is, each drive part **580** may extend between a respective drive link **574** and the respective vane drive member **562**.

(63) The vane drive member **562** may be connected to the drive link **574** and may be driven according to the operation of the drive link **574**.

(64) Meanwhile, the hygiene management device **1000** may include a first detector **354** and a second detector **356**. A first detector **354** may detect the opening/closing of the entrance door **14**, and the second detector **356** may detect whether a user is present at a preset position in the space **12** of the entrance hall **10**. The second detector **356** may preferably detect whether the user is present on the footrest **100**.

(65) The first detector **354** may be mounted to the entrance door **14**, and for example, may be embodied as a reed switch or a push-pull switch.

(66) The second detector **356** may be mounted to the front surface part of the function module **350**. However, the installation location of the second detector **356** may be any location in which a user on the footrest **100** can be detected. For example, the second detector **356** may be embodied as an infrared sensor, an ultrasonic sensor, or a laser distance sensor.

(67) Alternatively, the second detector **356** may be mounted to the footrest **100** and may detect whether a user stands on the footrest **100**. Accordingly, the second detector **356** may be embodied as a weight sensor or a piezoelectric sensor.

(68) The first and second detectors **354** and **356** may transmit results respectively detected thereby

to the controller **900**, which is described below.

(69) FIG. **10** is a block diagram of components connected to the controller which controls the hygiene management device of the present disclosure, and FIGS. **11** to **14** are the flowcharts illustrating the control method of the hygiene management device according to the embodiments of the present disclosure.

(70) Referring to FIGS. **10** and **11**, the operation of the hygiene management device **1000** according to the opening and closing of the entrance door **14** will be described.

(71) In a state in which the hygiene management device **1000** of the present disclosure is installed in the space **12** of the entrance hall **10**, as shown in FIG. **11**, the hygiene management device **1000** may initially maintain a standby state at **S101**. In the standby state, the air discharger **500** may be preset in a preset initial direction, and preferably, may be preset to be directed toward the center of the space **12** of the entrance hall **10**. Additionally, in the standby state, the air discharger **500** does not discharge air.

(72) In the standby state, when the entrance door **14** is opened, the first detector **354** may detect the opening of the entrance door **14** at **S103**. The detected result may be transmitted to the controller **900**.

(73) Accordingly, the controller **900** may drive the lift assembly **520** and may tilt the air discharger **500** such that the lower surface part of the air discharger **500** is directed toward the entrance door **14** at **S105**. That is, the direction of air discharged by the air discharger **500** may be directed toward the entrance door **14**.

(74) In addition, the controller **900** may drive the fan assembly **300** such that air is discharged by the air discharger **500** at **S107**.

(75) This is intended for the air discharger **500** to discharge air toward the entrance door **14** such that external air is prevented from being introduced into the entrance space **12** through the opened entrance door **14**. In this case, in the state in which the air discharger **500** is tilted, the fan assembly **300** may be driven to discharge air with a relatively higher wind pressure.

(76) FIG. **9B** illustrates an example in which the lower surface of the air discharger **500** is directed slantingly toward the entrance door **14** by tilting the air discharger **500** toward the entrance door **14**.

(77) FIG. **9A** illustrates that the lower surface of the air discharger **500** is directed downward, and FIG. **9B** illustrates that the lower surface of the air discharger **500** is directed rightward. For example, in FIG. **9B**, the height of the air discharger **500** at a side at which the first lift **521** is located is decreased, and the height of the air discharger **500** at a side at which the second lift **521'** is located is relatively increased.

(78) Accordingly, the air discharger **500** may discharge air slantingly toward the entrance door **14**. When the air discharger **500** operates in the opposite direction, the air discharger **500** may be restored to a present initial position from the tilted state of the air discharger **500**.

(79) Here, the tilting of the air discharger **500** at **S105** and the discharging of air at **S107** may be performed simultaneously or within a preset time difference. Alternatively, in another embodiment, the discharging of air may be performed before the tilting.

(80) Later, when the entrance door **14** is closed, the first detector **354** may detect the closing of the entrance door **14** at **S109**. The detected result may be transmitted to the controller **900**.

(81) Accordingly, the controller **900** may drive the lift assembly **520** such that the air discharger **500** tilted toward the entrance door **14** is restored to an initial position at **S111**. At the initial position, the direction of air discharged by the air discharger **500** may be directed toward the center of the space **12** of the entrance hall **10**.

(82) When the air discharger **500** is tilted to the initial position, whether to stop the air discharge may be determined according to whether there is a user in the space **12** of the entrance hall **10**.

(83) Referring to FIGS. **10** and **12**, the operation of the hygiene management device **1000** of the present disclosure when a user stands on the footrest **100** is described.

(84) The hygiene management device **1000** may maintain a standby state at **S201**, and when a user stands on the footrest **100**, the second detector **356** may detect the user on the footrest **100** at **S203**. The detected result may be transmitted to the controller **900**.

(85) Accordingly, the controller **900** may drive the fan assembly **300** such that the air discharger **500** discharges air toward the user at **S205**. This is intended to remove contaminants attached to the user's body and clothes by the air discharge.

(86) The air discharge may be performed for a preset period of time, and when the preset period of time elapses at **S207**, the controller **900** may stop the fan assembly **300** and the air discharge may stop at **S209**.

(87) However, before the preset period of time elapses, when the user descends from the footrest **100** (i.e., the user steps off the footrest **100**), the second detector **356** may detect this at **S211**, and the controller **900** may stop the fan assembly **300** such that the air discharge stops at **S209**.

(88) As described above, when the fan **316** operates due to the operation of the fan assembly **300**, an air flow may be generated. Hereinafter, the introduction and discharging of air according to the air flow will be described.

(89) When the fan assembly **300** is operated by the controller **900**, the fan **316** may rotate. Due to the rotation of the fan **316**, air may be introduced into the footrest **100** through the multiple introduction holes formed in the footrest **100**. In FIG. 15, an air flow for air introduction and discharge is illustrated properly.

(90) The introduced air may be introduced through the first, second, and third filters **221**, **223**, and **225** of the air management module **200** into the fan assembly **300**. The air introduced into the fan assembly **300** may flow due to the rotation of the fan **316**, and may be introduced into the duct **400**.

(91) The air transferred to the first duct **410** of the duct **400** may flow to the second duct **420** and may be transferred sequentially to the first outlet **422** and the second outlet **424** of the second duct **420**. The vanes **560** and **560'** may be located at the first outlet **422** and the second outlet **424**, respectively, and thus the flowing air may be introduced into the first outlet **422** and the second outlet **424** by being guided by the vanes **560** and **560'**, respectively.

(92) In addition, the multiple vanes may be disposed at each of the first outlet **422** and the second outlet **424** such that the vanes have a predetermined height difference therebetween and are spaced apart at predetermined intervals from each other, so air may be spread relatively evenly in the first outlet **422** and the second outlet **424** to flow.

(93) For the operation of the vane **560**, the drive link **574** may be reciprocated rectilinearly by the vane motor **570**. Due to the rectilinear reciprocation of the drive link **574**, the vane drive member **562** may move and thus the vane **560** may rotate at a predetermined angle relative to the vane rotation center member **558** (e.g., via the drive part **580**). Due to such rotation of the vane **560**, air discharge direction may be changed.

(94) In FIG. 18A, a state in which the drive link **574** is pulled toward the vane motor **570** is illustrated. In this state, air may be guided along the vane **560** and may be discharged toward the center of the lower side of the air discharger **500**. This is applied to the discharge hole corresponding to the first outlet **422** and the discharge hole corresponding to the second outlet **424**.

(95) In FIG. 18B, a state in which the drive link **574** is moved in a direction away from the vane motor **570** is illustrated. In this case, air may be discharged to the lower side of the air discharger **500**.

(96) Referring to FIGS. 10 and 13, when a user goes into the entrance space **12** through the entrance door **14** and stands on the footrest **100**, the operation of the hygiene management device **1000** of the present disclosure is described.

(97) When the entrance door **14** is opened such that a user goes into the entrance space **12** from the outside while the hygiene management device **1000** maintains a standby state at **S301**, the first detector **354** may detect the opening of the entrance door **14** at **S303**. The detected result may be transmitted to the controller **900**.

(98) Accordingly, the controller **900** may drive the lift assembly **520** and may tilt the air discharger **500** such that the lower surface of the air discharger **500** is directed toward the entrance door **14** at **S305**. Additionally, the controller **900** may enable the air discharger **500** tilted by driving the fan assembly **300** to discharge air toward the entrance door at **S307**.

(99) FIG. **16** illustrates the performance of the tilting and air discharging of the air discharger **500** when a user comes into the entrance space through the entrance door **14**. In the process in which the user comes into the entrance space through the entrance door **14**, external air may be prevented from being introduced into the entrance space **12** through the opened entrance door **14** by discharging air through the air discharger **500** toward the entrance door **14**.

(100) Next, when the entrance door **14** is closed after the user comes into the space **12** of the entrance hall **10**, the first detector **354** may detect the closing of the entrance door **14** at **S309**. The detected result may be transmitted to the controller **900**.

(101) Accordingly, the controller **900** may stop the fan assembly **300** such that air discharge stops at **S311**. Additionally, the controller **900** may drive the lift assembly **520** such that the air discharger **500** tilted toward the entrance door **14** is restored to an initial position at **S313**.

(102) Later, when a user is located on the footrest **100**, the second detector **356** may detect this at **S315**. The result detected by the second detector **356** may be transmitted to the controller **900** (e.g., as shown in FIG. **9A**).

(103) Accordingly, the controller **900** may drive the fan assembly **300** such that the air discharger **500** discharges air toward a user at **S317**. In this case, due to the operation of the fan assembly **300**, the footrest **100** may introduce air in the entrance space **12** thereto through the multiple introduction holes.

(104) The air discharge may be performed for a preset period of time, and when the preset period of time elapses at **S319**, the controller **900** may stop the fan assembly **300** and stop the air discharge at **S321**.

(105) However, before the preset period of time elapses, when the user descends from the footrest **100** (i.e., the user steps off of the footrest **100**), the second detector **356** may detect this at **S323**, and the controller **900** may stop the operation of the fan assembly **300** to stop air discharge at **S321**.

(106) Referring to FIGS. **10** and **14**, the operation of the hygiene management device **1000** of the present disclosure when a user goes out to the outside through the entrance door **14** is described.

(107) In a state in which the hygiene management device **1000** is in a standby state at **S401**, when a user coming out through the inside door **16** stands on the footrest **100**, the second detector **356** may detect the user on the footrest **100** at **S403** and may transmit the detected result to the controller **900**.

(108) Accordingly, the controller **900** may drive the fan assembly **300** such that the air discharger **500** discharges air toward the user at **S405**.

(109) The air discharge may be performed for a preset period of time, and when the preset period of time elapses at **S407**, the controller **900** may stop the fan assembly **300** and may stop the air discharge at **S409**.

(110) When the entrance door **14** is opened such that the user goes out to the outside from the entrance space, the first detector **354** may detect the opening of the entrance door **14** at **S411** and may transmit the detected result to the controller **900**.

(111) Accordingly, the controller **900** may drive the lift assembly **520** and tilt the air discharger **500** toward the entrance door **14** at **S413**. Additionally, the controller **900** may drive the fan assembly **300** such that the air discharger **500** discharges air toward the entrance door **14** at **S415**.

(112) The tilting of the air discharger **500** at **S413** and the discharging of air at **S415** may be performed simultaneously or within a preset time difference.

(113) FIG. **17** illustrates an example in which the tilting (e.g., toward the entrance door **14**) and air discharge of the air discharger **500** are performed when a user goes out to the outside through the entrance door **14**. In the process in which the user goes out to the outside through the entrance door

14, external air may be prevented from being introduced into the entrance space **12** through the opened entrance door **14** by discharging air toward the entrance door **14**.

(114) Later, when the entrance door **14** is closed after the user goes out to the outside, the first detector **354** may detect the closing of the entrance door **14** at **S417**. The detected result may be transmitted to the controller **900**.

(115) Accordingly, the controller **900** may stop the operation of the fan assembly **300** such that the air discharger **500** stops the air discharge at **S419**. Additionally, the controller **900** may drive the lift assembly **520** such that the air discharger **500** tilted toward the entrance door **14** is tilted to an initial position to be restored thereto at **S421**, as shown in FIG. **9A**.

(116) Various embodiments described herein may be implemented in a computer-readable medium using, for example, software, hardware, or some combination thereof. For example, the embodiments described herein may be implemented within one or more of Application Specific Integrated Circuits (ASICs), Digital Signal Processors (DSPs), Digital Signal Processing Devices (DSPDs), Programmable Logic Devices (PLDs), Field Programmable Gate Arrays (FPGAs), processors, controllers, micro-controllers, microprocessors, other electronic units designed to perform the functions described herein, or a selective combination thereof. In some cases, such embodiments are implemented by the controller. That is, the controller is a hardware-embedded processor executing the appropriate algorithms (e.g., flowcharts) for performing the described functions and thus has sufficient structure. Also, the embodiments such as procedures and functions may be implemented together with separate software modules each of which performs at least one of functions and operations. The software codes can be implemented with a software application written in any suitable programming language. Also, the software codes can be stored in the memory and executed by the controller, thus making the controller a type of special purpose controller specifically configured to carry out the described functions and algorithms. Thus, the components shown in the drawings have sufficient structure to implement the appropriate algorithms for performing the described functions.

(117) In the above, although all components constituting the hygiene management device according to the embodiment of the present disclosure are described as being integrally combined or being operated in combination, the present disclosure is not necessarily limited to this embodiment. That is, within the scope of the present disclosure, at least two of all the components may operate by being selectively combined with each other. In addition, terms such as “include”, “comprise”, or “have” described above mean that the associated components may be inherent unless otherwise stated, so the terms should be construed not to exclude other components but to further include the other components.

Claims

1. A hygiene management device for an entrance hall, the device comprising: a first detector configured to detect an opening/closing of an entrance door located in the entrance hall; an air discharger located in the entrance hall and configured to discharge air in a downward direction toward a space; a fan assembly configured to generate a flow of air toward the air discharger to allow the air discharger to discharge air to the space; a duct connecting the fan assembly with the air discharger; a lift assembly configured to tilt the air discharger; a plurality of vanes configured to control a direction of air discharged through the air discharger; and a controller configured to drive the fan assembly and the lift assembly, in response to the first detector detecting the opening of the entrance door, the controller being configured to: drive the lift assembly such that the air discharger is tilted, and drive the fan assembly such that the tilted air discharger discharges air, wherein upper portions of the plurality of vanes are disposed in the duct and lower portions of the plurality of vanes are disposed in the air discharger, wherein the lift assembly includes: a first lift mounted at a first side of the air discharger; a second lift mounted at a second side of the air discharger opposite

to the first side of the air discharger; and at least one lift motor configured to drive the first lift and the second lift, and wherein the first lift is configured to operate to tilt the air discharger such that the first side of the air discharger is lowered and the air discharger is angled toward the entrance door.

2. The hygiene management device of claim 1, wherein the plurality of vanes includes a first plurality of vanes and a second plurality of vanes wherein the second plurality of vanes are located downstream in the duct with respect to the first plurality of vanes, and wherein a length of at least one upper portion of the plurality of second vanes is larger than a length of at least one upper portion of the first plurality of vanes.
3. The hygiene management device of claim 1, wherein the controller is configured to stop an operation of the fan assembly such that the discharging of air stops in response to a preset period of time elapsing after the discharging of air.
4. The hygiene management device of claim 3, further comprising: a footrest positioned at a bottom of the space of the entrance hall; and a second detector configured to detect whether a user is present on the footrest, wherein the controller is configured to stop the operation of the fan assembly such that the discharging of air stops in response to the user not being detected on the footrest by the second detector before the preset period of time elapses.
5. The hygiene management device of claim 4, wherein the footrest includes a plurality of introduction holes, and wherein the plurality of introduction holes are configured to receive air discharged from the air discharger.
6. The hygiene management device of claim 1, wherein the controller is configured to, while the air discharger is tilted, in response to the closing of the entrance door being detected by the first detector, drive the lift assembly to tilt the tilted air discharger to an initial position, and wherein in the initial position, the air discharger faces a center of the space of the entrance hall.
7. The hygiene management device of claim 1, further comprising: a footrest positioned at a bottom of the space of the entrance hall; and a second detector configured to detect whether a user is present on the footrest, wherein the controller is configured to drive the fan assembly such that the air discharger discharges air toward the user in response to the second detector detecting the user is on the footrest.
8. The hygiene management device of claim 2, wherein the first plurality of vanes includes a first vane disposed at the first side of the air discharger; and wherein the second plurality of vanes includes a second vane disposed at the second side of the air discharger spaced from the first vane, and wherein the first vane and the second vane are configured to receive air from the fan assembly and to discharge air in a preset direction.
9. The hygiene management device of claim 8, wherein the first vane is disposed closer to the fan assembly than the second vane, and wherein the second vane is disposed higher with respect to a vertical direction than the first vane.
10. The hygiene management device of claim 8, wherein the first vane has a longer length than the second vane.
11. The hygiene management device of claim 1 wherein the duct comprises: a first duct disposed in a vertical direction, and a second duct that connects to the first duct and is disposed in a horizontal direction, wherein the first duct is disposed closer to the fan assembly than the second duct, and wherein the second duct is disposed closer to the air discharger than the first duct.
12. The hygiene management device of claim 11, wherein the upper portions of the plurality of vanes are disposed in the second duct.
13. The hygiene management device of claim 11, wherein the first lift and second lift are located under the second duct with respect to the vertical direction.
14. The hygiene management device of claim 1, wherein each of the first lift and the second lift is configured to move in a vertical direction.
15. The hygiene management device of claim 1, wherein the at least one lift motor comprises: a

first lift motor configured to drive the first lift; and a second lift motor configured to drive the second lift.

16. The hygiene management device of claim 15, wherein the first lift motor and the second lift motor are each located on a bottom surface of the duct.
